10 PROPERTIES OF WAVES

Talking to someone using a mobile phone is something most of us do several times a day. The technology that had to be developed for this to happen was based on a thorough understanding of the properties of waves.

In this chapter you will learn about different types of waves and their properties (characteristics).



▲ Figure 10.1 Using microwaves to communicate

LEARNING OBJECTIVES

- Explain the difference between longitudinal and transverse waves
- Know the definitions of amplitude, wavefront, frequency, wavelength and period of a wave
- Know that waves transfer energy and information without transferring matter
- Know and use the relationship between the speed, frequency and wavelength of a wave:

  wave speed = frequency × wavelength

  $v = f \times \lambda$
- Use the relationship between frequency and time period:

  $\text{frequency} = \frac{1}{\text{time period}}$

  $f = \frac{1}{T}$
- Use the above relationships in different contexts including sound waves and electromagnetic waves
- Explain that all waves can be reflected and refracted
- Explain why there is a change in the observed frequency and wavelength of a wave when its source is moving relative to an observer, and that this is known as the Doppler effect

UNITS

In this unit, you will need to use degrees (°) as the unit of angle, hertz (Hz) as the unit of frequency, metre (m) as the unit of length, metre per second (m/s) as the unit of speed and second (s) as the unit of time.

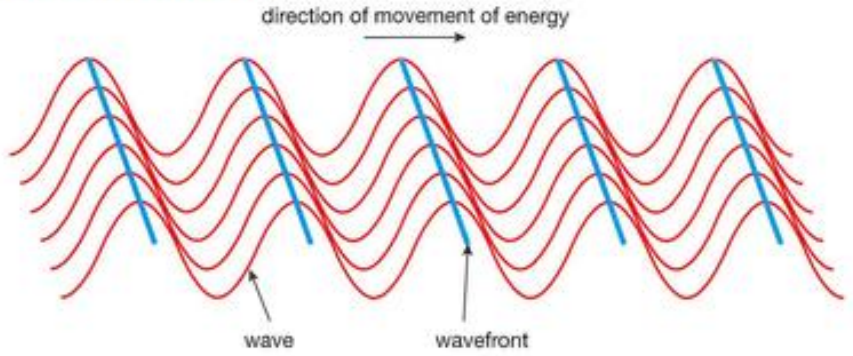
WHAT ARE WAVES?

Waves are a way of transferring energy from place to place. As we can see in Figure 10.1 we often use them to transfer information. All these transfers take place with no matter being transferred.



▲ Figure 10.2 Waves are produced if we drop a stone into a pond. The circular wavefronts spread out from the point of impact, carrying energy in all directions, but the water in the pond does not move from the centre to the edges.

WHAT ARE WAVEFRONTS?

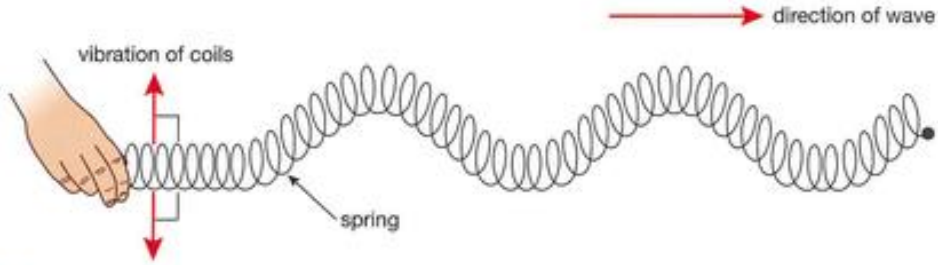


▲ Figure 10.3 Wavefronts are created by overlapping lots of different waves. A wavefront is a line where all the vibrations are in phase and the same distance from the source.

TRANSVERSE WAVES

Waves can be produced in ropes and springs. If you move one end of a spring from side to side you will see waves travelling through it. The energy carried by these waves moves along the spring from one end to the other, but if you look closely you can see that the coils of the spring are vibrating (shaking) across the direction in which the energy is moving. This is an example of a **transverse wave**.

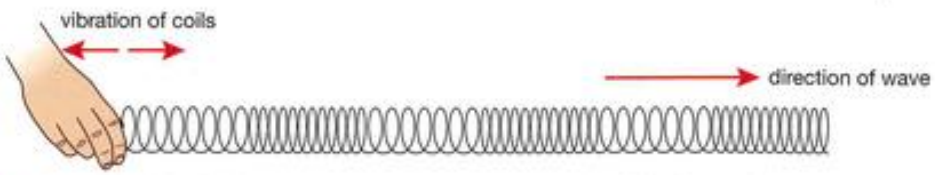
A transverse wave is one that vibrates, or oscillates, at right angles to the direction in which the energy or wave is moving. Examples of transverse waves include light waves and waves travelling on the surface of water.



▲ Figure 10.4 A transverse wave vibrates at right angles to the direction in which the wave is moving.

LONGITUDINAL WAVES

If you push and pull the end of a spring in a direction parallel to its axis, you can again see energy travelling along it. This time however the coils of the spring are vibrating in directions that are along its length. This is an example of a **longitudinal wave**.

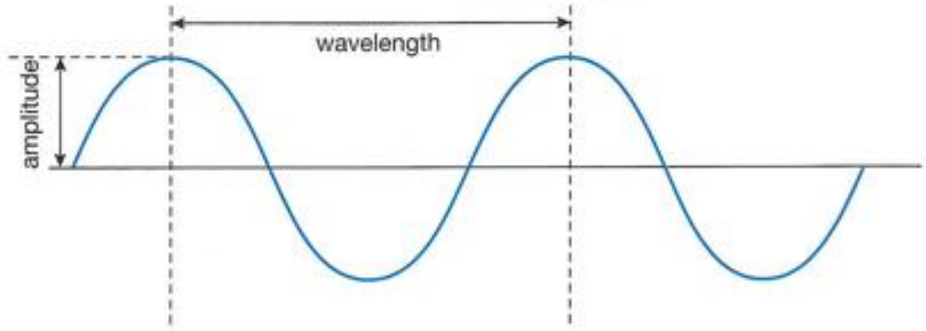


▲ Figure 10.5 A longitudinal wave vibrates along the direction in which the wave is travelling.

A longitudinal wave is one in which the vibrations, or oscillations, are along the direction in which the energy or wave is moving. Examples of longitudinal waves include sound waves.

DESCRIBING WAVES

When a wave moves through a substance, its particles will move from their equilibrium (resting position). The maximum movement of particles from their resting or equilibrium position is called its **amplitude** (A).



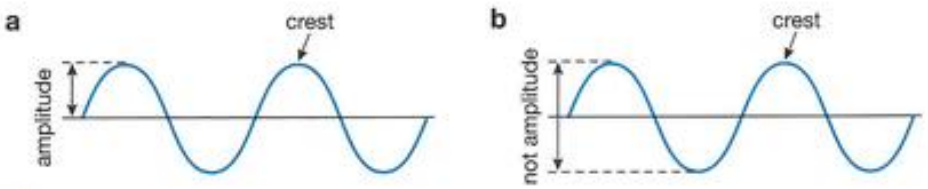
▲ Figure 10.6 A wave has amplitude and wavelength.

The distance between a particular point on a wave and the same point on the next wave (for example, from crest to crest) is called the **wavelength** (λ).

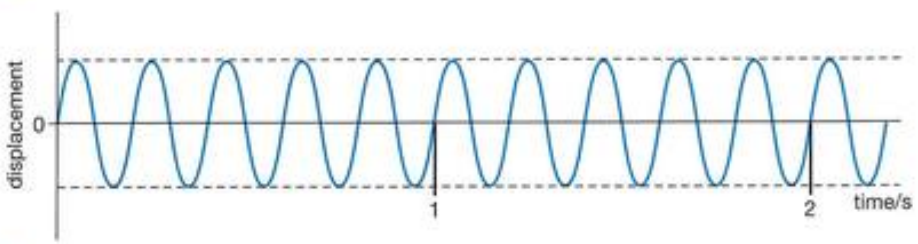
If the source that is creating a wave vibrates quickly it will produce a large number of waves each second. If it vibrates more slowly it will produce fewer waves each second. The number of waves produced each second by a source, or the number passing a particular point each second, is called the frequency of the wave (f). Frequency is measured in hertz (Hz). A wave source that produces five complete waves each second has a frequency of 5 Hz.

KEY POINT

λ is the Greek letter lambda and is the usual symbol for wavelength.



▲ Figure 10.7 The amplitude of a wave is as shown in **a**, and not as in **b**.



▲ Figure 10.8 This graph shows a wave with a frequency of 5 Hz.

The time it takes for a source to produce one wave is called the **time period** of the wave (T). It is related to the frequency (f) of a wave by the equation:

$$\text{frequency, } f \text{ (Hz)} = \frac{1}{\text{time period, } T \text{ (s)}}$$

$$f = \frac{1}{T}$$

This equation can also be written as

$$T = \frac{1}{f}$$

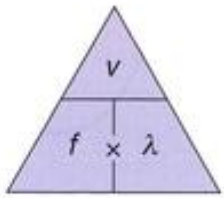
EXAMPLE 1

Calculate the period of a wave with a frequency of 200 Hz.

$$T = \frac{1}{f}$$

$$= \frac{1}{200\ \text{Hz}}$$

$$= 0.005 \text{ or } 5\ \text{ms}\ (1000\ \text{ms} = 1\ \text{s})$$



▲ Figure 10.9 You can use the triangle method for rearranging equations like $v = f \times \lambda$.

HINT

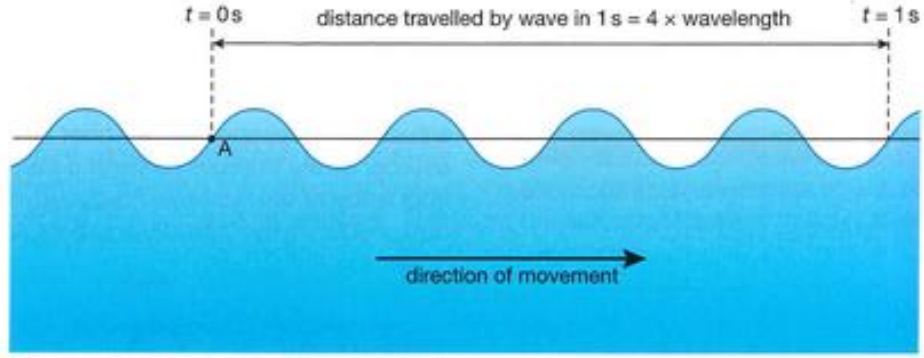
If an examination question asks you to write out the equation for calculating wave speed, wavelength or frequency, always give the actual equation such as $v = f \times \lambda$. You may not be awarded a mark if you just draw the triangle.

THE WAVE EQUATION

There is a relationship between the wavelength (λ), the frequency (f) and the wave speed (v) that is true for all waves:

wave speed, v (m/s) = frequency, f (Hz) × wavelength, λ (m)

$$v = f \times \lambda$$



▲ Figure 10.10 A wave with a frequency of 4 Hz

Imagine that you have created water waves with a frequency of 4 Hz. This means that four waves will pass a particular point each second. If the wavelength of the waves is 3 m, then the waves travel 12 m each second. The speed of the waves is therefore 12 m/s.

$$v = f \times \lambda$$

$$= 4\ \text{Hz} \times 3\ \text{m}$$

$$= 12\ \text{m/s}$$

EXAMPLE 2

A **tuning fork** creates sound waves with a frequency of 170 Hz. If the speed of sound in air is 340 m/s, calculate the wavelength of the sound waves.

$$v = f \times \lambda$$

So $$\lambda = \frac{v}{f}$$

$$= \frac{340\ \text{m/s}}{170\ \text{Hz}}$$

$$= 2\ \text{m}$$

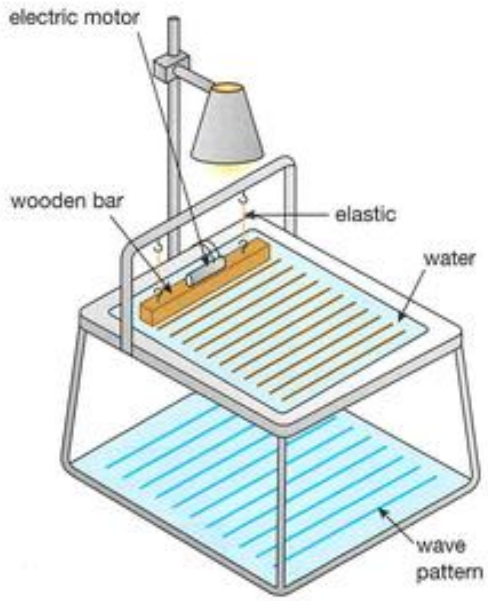


▲ Figure 10.11 A tuning fork

THE RIPPLE TANK

We can study the behaviour of water waves using a ripple tank.

When the motor is turned on, the wooden bar vibrates creating a series of ripples or wavefronts on the surface of the water. A light placed above the tank creates patterns of the water waves on the floor. By observing the patterns we can see how the water waves are behaving.

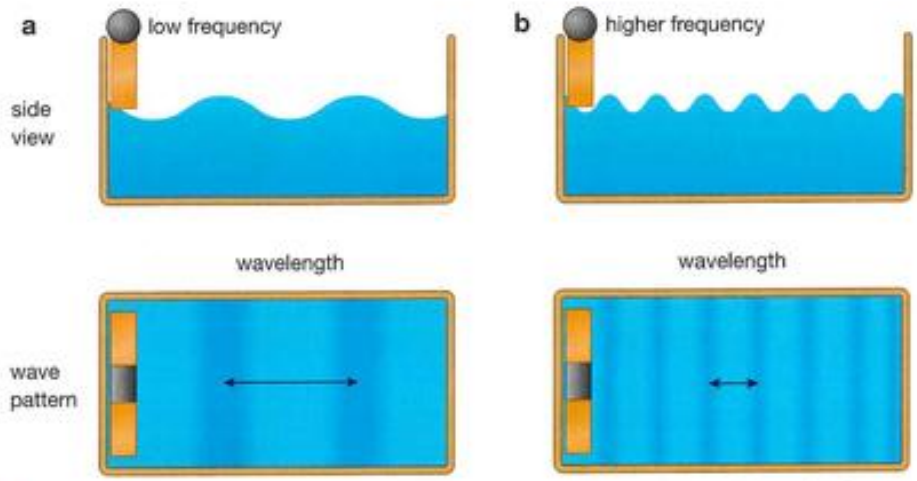


▲ Figure 10.12 The light shines through the water and we can see the patterns of the waves.

WAVELENGTH AND FREQUENCY

The motor can be adjusted to produce a small number of waves each second. The frequency of the waves is small and the pattern shows that the waves have a long wavelength.

At higher frequencies, the water waves have shorter wavelengths. The speed of the waves does not change.



▲ Figure 10.13 When the frequency of the waves is low, the wavelength is long. When the frequency is higher, the wavelength is shorter.

REFLECTION

All waves can be reflected. If they hit a straight or flat barrier, the angle at which they leave the barrier surface is equal to the angle at which they meet the surface – that is, the waves are reflected from the barrier at the same angle as they strike it. This is described by the 'Law of Reflection' which states that:

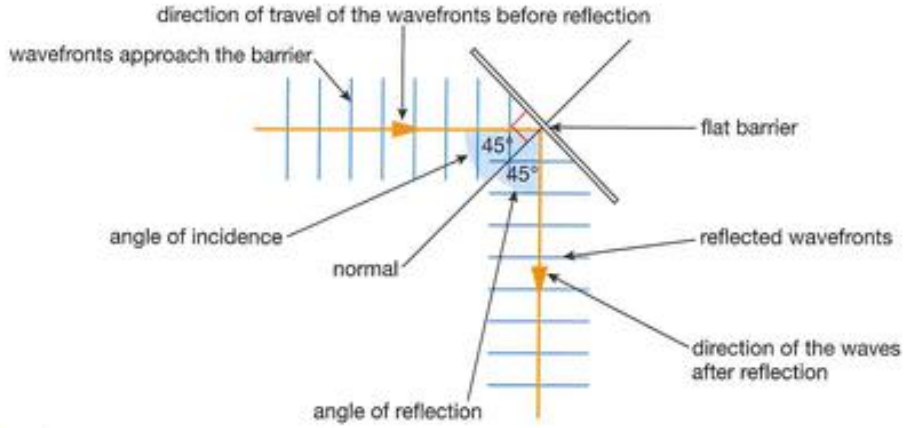
*The **angle of incidence** is equal to the **angle of reflection**.*

KEY POINT

A normal is a line drawn at right angles to a surface.

The angle of incidence is the angle between the direction of the waves as they approach the barrier and the normal.

The angle of reflection is the angle between the direction of the waves after striking the barrier and the normal.

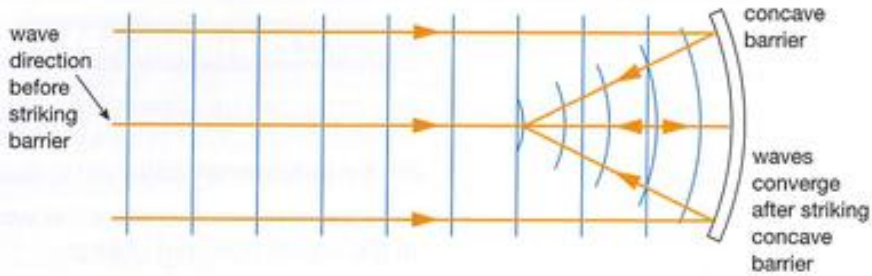


▲ Figure 10.14 Waves striking a flat barrier are reflected. The angle at which they strike the barrier is the same as the angle at which they are reflected.

EXTENSION WORK

Although you will not be asked this in your exam, it is interesting to see how waves are reflected from curved surfaces.

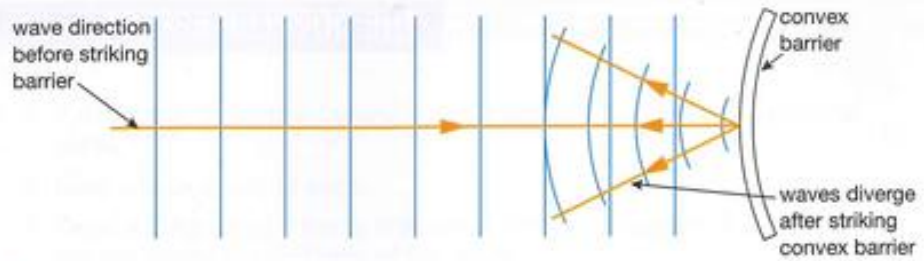
When the waves strike a concave barrier, they are made to converge (come together).



▲ Figure 10.15 Waves striking a concave barrier are reflected and converge.



▲ Figure 10.16 Radio telescopes have concave reflecting dishes so that the signals from space reflect and converge onto a detector that is placed in front of the dish.



▲ Figure 10.17 Waves striking a convex barrier are reflected backwards and spread out.

When waves are reflected by a surface that is curved outwards (convex), they diverge (spread out).

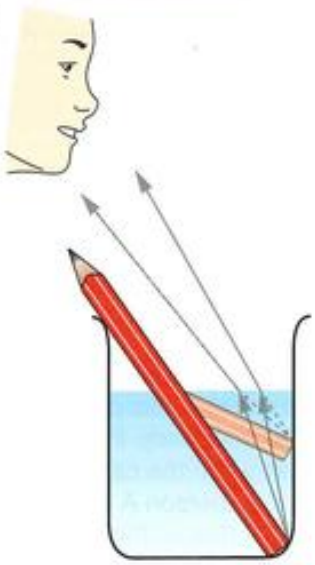


▲ Figure 10.18 The light waves reflected from this concave make-up mirror create a magnified image.

a



b



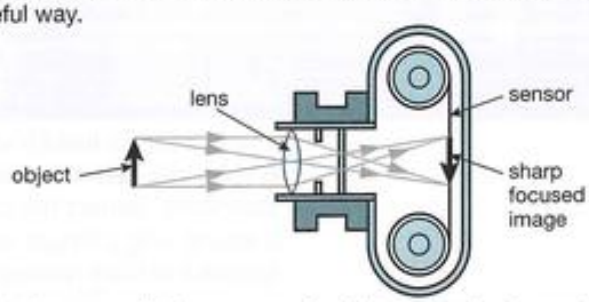
▲ Figure 10.19 **a** The pencil seems to bend at the air/water boundary. **b** Why the pencil appears to be bent. Rays of light are refracted at the water surface.

REFRACTION

The pencil in Figure 10.19 is straight but it seems to bend at the surface of the water. This happens because light waves in water travel more slowly than light waves in air. This change in speed as they leave the water causes the light waves to change direction. This change in direction is called refraction. All waves – light waves, sound waves, water waves – can be refracted.

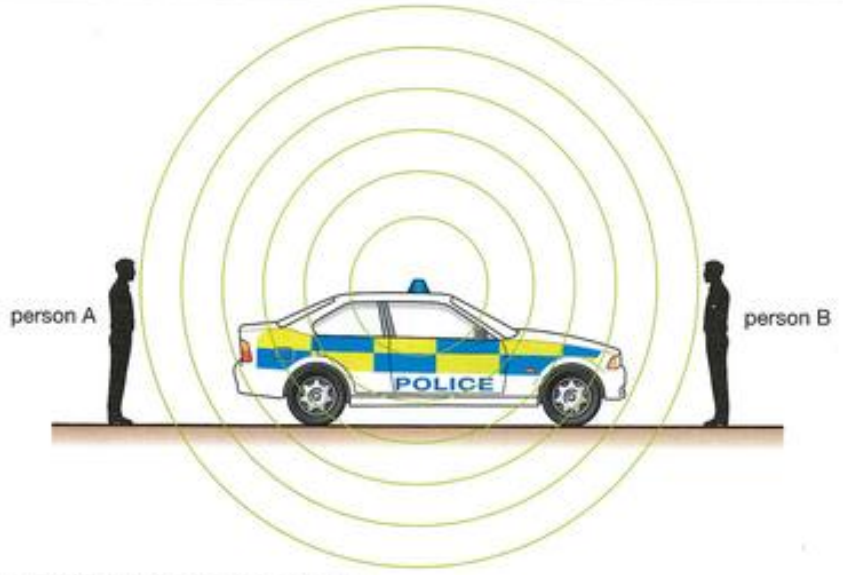
EXTENSION WORK

Many optical instruments such as microscopes, telescopes and cameras use specially shaped pieces of glass or plastic (called lenses) to bend or refract light waves in a useful way.



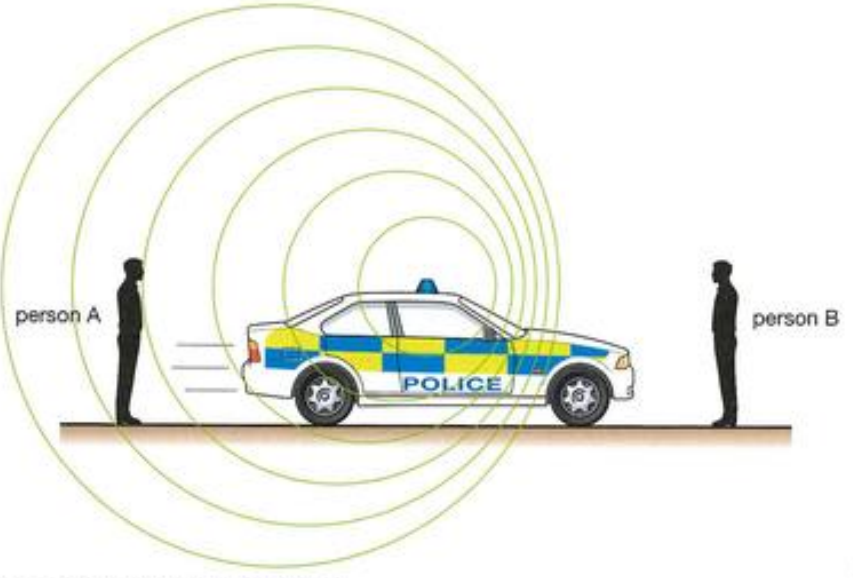
▲ Figure 10.20 In this camera, light waves are refracted by a glass lens to create a sharp image on the sensor or film. Refraction occurs because light travels more slowly in glass than in air.

THE DOPPLER EFFECT



▲ Figure 10.21 A stationary source of sound

When a car is not moving the sound waves we receive from its engine or from its horn arrive as a series of equally (evenly) spaced wavefronts. People in front of and behind the car hear sound of the same frequency and wavelength.



▲ Figure 10.22 A moving source of sound

If the car is moving, the wavefronts are no longer evenly spaced. Ahead of the car the wavefronts will be compressed as the car is moving in this direction. The waves will have a shorter wavelength and a higher frequency. Person B therefore hears a sound that has a higher **pitch** than when the car was stationary. Behind the car the waves are stretched out so person A hears a sound with a longer wavelength and lower frequency – that is, the pitch appears to have decreased.

These apparent changes in frequency, which occur when a source of waves is moving, is called the **Doppler effect** and is a property of all waves.

CHAPTER QUESTIONS

More questions on wave properties can be found at the end of Unit 3 on page 130.

SKILLS CRITICAL THINKING

6

1 a Explain the difference between a transverse wave and a longitudinal wave.

5

b Give one example of each.

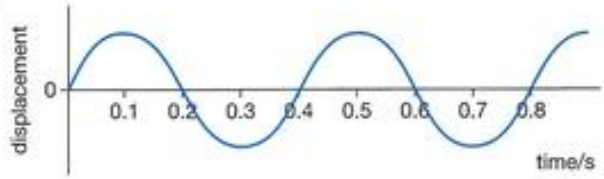
SKILLS INTERPRETATION

c Draw a diagram of a transverse wave. On your diagram, mark the wavelength and amplitude of the wave.

SKILLS ANALYSIS, PROBLEM SOLVING

7

2 The diagram below shows the displacement of water as a wave travels through it.



From the diagram calculate:

a the period of the wave

b the frequency of the wave.

SKILLS PROBLEM SOLVING

3 The speed of sound in water is approximately 1500 m/s.

a What is the frequency of a sound wave with a wavelength of 1.5 m?

8

b What is the period of this wave?

SKILLS CRITICAL THINKING

6

4 a Explain why the sound produced by the horn of an approaching car seems to have a higher frequency than one that is stationary.

SKILLS REASONING

b If the same car approached at a much higher speed how would this affect the frequency of the sound heard?

c Describe the frequency of the sound heard by the observer if the car is moving away at high speed.

6

5 Explain why this hunter should not aim at the fish he can see.

